



МИНОБРНАУКИ РОССИИ

Федеральное государственное бюджетное образовательное учреждение
высшего образования

«МИРЭА – Российский технологический университет»

РТУ МИРЭА

Институт информационных технологий (ИИТ)

Кафедра цифровой трансформации (ЦТ)

ОТЧЕТ ПО ПРАКТИЧЕСКОЙ РАБОТЕ

по дисциплине «Проектирование баз данных»

Практическая работа № 6

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Цель работы:

Целью данной практической работы является осуществление корректного перехода от логической реляционной модели данных к оптимизированному физическому DDL-скрипту (Data Definition Language). Студентам предстоит транслировать абстрактные структуры под конкретные архитектурные особенности, механизмы выделения памяти и строгие стандарты СУБД PostgreSQL.

Выбранная предметная область

В данной работе проектирование базы данных будет вестись для типового учреждения культуры - «Городская библиотека».

Ход выполнения работы

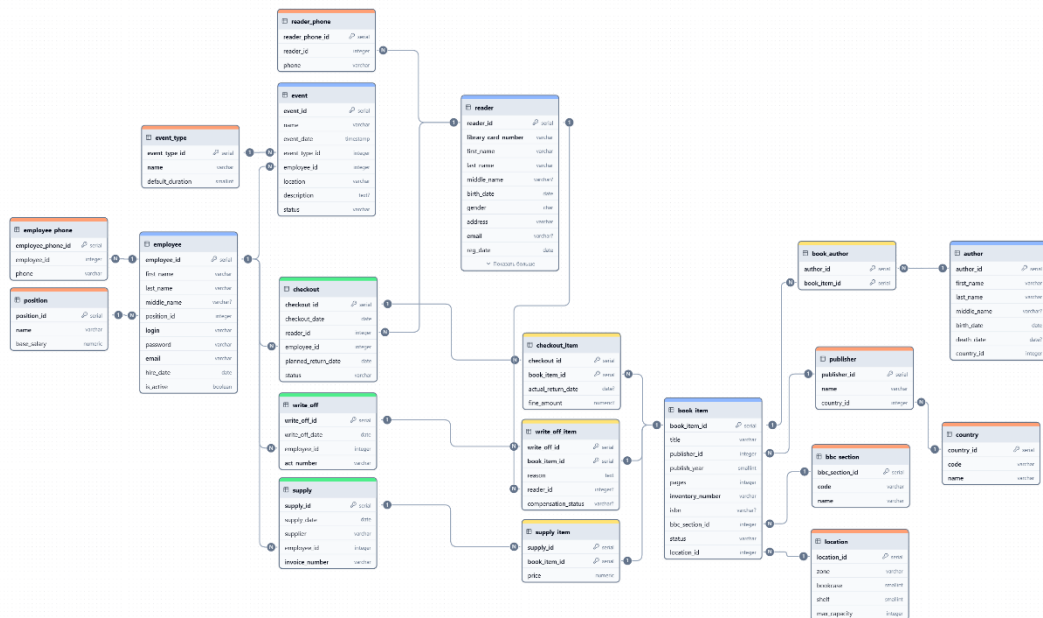


Рисунок 1. Готовая диаграмма

Листинг 1. Итоговый SQL-скрипт

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-- УРОВЕНЬ 1. НЕЗАВИСИМЫЕ СПРАВОЧНИКИ
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CREATE TABLE country (
    country_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    code varchar(3) NOT NULL UNIQUE,
    name varchar(100) NOT NULL UNIQUE
);

CREATE TABLE position (
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    position_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(100) NOT NULL UNIQUE,
    base_salary numeric(10,2) NOT NULL
);

CREATE TABLE bbc_section (
    bbc_section_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    code varchar(20) NOT NULL UNIQUE,
    name varchar(200) NOT NULL UNIQUE
);

CREATE TABLE event_type (
    event_type_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(100) NOT NULL UNIQUE,
    default_duration smallint NOT NULL DEFAULT 60
);

CREATE TABLE location (
    location_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    zone varchar(50) NOT NULL,
    bookcase smallint NOT NULL,
    shelf smallint NOT NULL,
    max_capacity integer NOT NULL DEFAULT 100
);

--
-- УРОВЕНЬ 2. ЗАВИСИМЫЕ СПРАВОЧНИКИ
--

CREATE TABLE publisher (
    publisher_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(200) NOT NULL UNIQUE,
    country_id integer NOT NULL
);

--
-- УРОВЕНЬ 3. КЛЮЧЕВЫЕ ОБЪЕКТЫ
--

CREATE TABLE employee (
    employee_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    first_name varchar(100) NOT NULL,
    last_name varchar(100) NOT NULL,
    middle_name varchar(100),
    position_id integer NOT NULL,
    login varchar(50) NOT NULL UNIQUE,
    password varchar(255) NOT NULL,
    email varchar(100) NOT NULL UNIQUE,
    hire_date date NOT NULL DEFAULT CURRENT_DATE,
    is_active boolean NOT NULL DEFAULT true
);

CREATE TABLE reader (
    reader_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    library_card_number varchar(20) NOT NULL UNIQUE,
    first_name varchar(100) NOT NULL,
    last_name varchar(100) NOT NULL,
    middle_name varchar(100),
    birth_date date NOT NULL,
    gender char(1) NOT NULL,

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address varchar(255) NOT NULL,
email varchar(100) UNIQUE,
reg_date date NOT NULL DEFAULT CURRENT_DATE,
is_active boolean NOT NULL DEFAULT true
);

CREATE TABLE book_item (
book_item_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
title varchar(255) NOT NULL,
publisher_id integer NOT NULL,
publish_year smallint NOT NULL,
pages integer NOT NULL,
inventory_number varchar(50) NOT NULL UNIQUE,
isbn varchar(20),
bbc_section_id integer NOT NULL,
status varchar(20) NOT NULL DEFAULT 'В наличии',
location_id integer NOT NULL
);

CREATE TABLE author (
author_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
first_name varchar(100) NOT NULL,
last_name varchar(100) NOT NULL,
middle_name varchar(100),
birth_date date NOT NULL,
death_date date,
country_id integer NOT NULL
);

--
-- УРОВЕНЬ 4. ДОКУМЕНТЫ И РЕГИСТРЫ
--

CREATE TABLE event (
event_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
name varchar(100) NOT NULL,
event_date timestamp NOT NULL,
event_type_id integer NOT NULL,
employee_id integer NOT NULL,
location varchar(100) NOT NULL,
description text,
status varchar(20) NOT NULL DEFAULT 'Планируется'
);

CREATE TABLE checkout (
checkout_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
checkout_date date NOT NULL DEFAULT CURRENT_DATE,
reader_id integer NOT NULL,
employee_id integer NOT NULL,
planned_return_date date NOT NULL DEFAULT (CURRENT_DATE + INTERVAL '30
days'),
status varchar(20) NOT NULL DEFAULT 'Активна'
);
COMMENT ON COLUMN checkout.status IS 'Статус выдачи: "Активна" или
"Закрыта"';

CREATE TABLE supply (
supply_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
supply_date date NOT NULL DEFAULT CURRENT_DATE,
supplier varchar(200) NOT NULL,

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    employee_id integer NOT NULL,
    invoice_number varchar(50) NOT NULL UNIQUE
);

CREATE TABLE write_off (
    write_off_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    write_off_date date NOT NULL DEFAULT CURRENT_DATE,
    employee_id integer NOT NULL,
    act_number varchar(50) NOT NULL UNIQUE
);
COMMENT ON TABLE write_off IS 'Акт списания ветхих, утерянных или испорченных книг. Содержит номер акта, дату и сотрудника, оформившего списание.';

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-- УРОВЕНЬ 5. ПРОМЕЖУТОЧНЫЕ ТАБЛИЦЫ (позиции документов и связи M:N)
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CREATE TABLE checkout_item (
    checkout_id integer NOT NULL,
    book_item_id integer NOT NULL,
    actual_return_date date,
    fine_amount numeric(10,2),
    PRIMARY KEY (checkout_id, book_item_id)
);
COMMENT ON TABLE checkout_item IS 'Список книг, выданных читателю в рамках одного чека, с информацией о возврате и штрафах';

COMMENT ON COLUMN checkout_item.actual_return_date IS 'Дата фактического возврата книги (NULL означает, что книга ещё на руках у читателя)';

CREATE TABLE supply_item (
    supply_id integer NOT NULL,
    book_item_id integer NOT NULL,
    price numeric(10,2) NOT NULL,
    PRIMARY KEY (supply_id, book_item_id)
);

CREATE TABLE write_off_item (
    write_off_id integer NOT NULL,
    book_item_id integer NOT NULL,
    reason text NOT NULL,
    reader_id integer,
    compensation_status varchar(20),
    PRIMARY KEY (write_off_id, book_item_id)
);

CREATE TABLE employee_phone (
    employee_phone_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    employee_id integer NOT NULL,
    phone varchar(20) NOT NULL
);

CREATE TABLE reader_phone (
    reader_phone_id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    reader_id integer NOT NULL,
    phone varchar(20) NOT NULL
);

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CREATE TABLE book_author (
    author_id integer NOT NULL,
    book_item_id integer NOT NULL,
    PRIMARY KEY (author_id, book_item_id)
);

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-- БЛОК ОГРАНИЧЕНИЙ ВНЕШНИХ КЛЮЧЕЙ (С ПРАВИЛАМИ ON DELETE / ON UPDATE)
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-- Защита справочников: объекты зависят от справочников (ON DELETE RESTRICT)
ALTER TABLE publisher ADD CONSTRAINT publisher_country_id_fk FOREIGN KEY
(country_id) REFERENCES country (country_id) ON DELETE RESTRICT;
ALTER TABLE author ADD CONSTRAINT author_country_id_fk FOREIGN KEY
(country_id) REFERENCES country (country_id) ON DELETE RESTRICT;
ALTER TABLE employee ADD CONSTRAINT employee_position_id_fk FOREIGN KEY
(position_id) REFERENCES position (position_id) ON DELETE RESTRICT;
ALTER TABLE event ADD CONSTRAINT event_event_type_id_fk FOREIGN KEY
(event_type_id) REFERENCES event_type (event_type_id) ON DELETE RESTRICT;
ALTER TABLE book_item ADD CONSTRAINT book_item_publisher_id_fk FOREIGN KEY
(publisher_id) REFERENCES publisher (publisher_id) ON DELETE RESTRICT;
ALTER TABLE book_item ADD CONSTRAINT book_item_bbc_section_id_fk FOREIGN KEY
(bbc_section_id) REFERENCES bbc_section (bbc_section_id) ON DELETE RESTRICT;
ALTER TABLE book_item ADD CONSTRAINT book_item_location_id_fk FOREIGN KEY
(location_id) REFERENCES location (location_id) ON DELETE RESTRICT;

-- Зависимые атрибуты объектов: телефоны исчезают вместе с владельцем (ON
DELETE CASCADE)
ALTER TABLE employee_phone ADD CONSTRAINT employee_phone_employee_id_fk
FOREIGN KEY (employee_id) REFERENCES employee (employee_id) ON DELETE
CASCADE;
ALTER TABLE reader_phone ADD CONSTRAINT reader_phone_reader_id_fk FOREIGN KEY
(reader_id) REFERENCES reader (reader_id) ON DELETE CASCADE;

-- Историческая безопасность: документы блокируют удаление участников сделок
(ON DELETE RESTRICT)
ALTER TABLE checkout ADD CONSTRAINT checkout_employee_id_fk FOREIGN KEY
(employee_id) REFERENCES employee (employee_id) ON DELETE RESTRICT;
ALTER TABLE checkout ADD CONSTRAINT checkout_reader_id_fk FOREIGN KEY
(reader_id) REFERENCES reader (reader_id) ON DELETE RESTRICT;
ALTER TABLE supply ADD CONSTRAINT supply_employee_id_fk FOREIGN KEY
(employee_id) REFERENCES employee (employee_id) ON DELETE RESTRICT;
ALTER TABLE write_off ADD CONSTRAINT write_off_employee_id_fk FOREIGN KEY
(employee_id) REFERENCES employee (employee_id) ON DELETE RESTRICT;
ALTER TABLE write_off_item ADD CONSTRAINT write_off_item_reader_id_fk FOREIGN
KEY (reader_id) REFERENCES reader (reader_id) ON DELETE RESTRICT;

-- Целостность документов: промежуточные таблицы жёстко зависят от документов
(ON DELETE CASCADE)
ALTER TABLE checkout_item ADD CONSTRAINT checkout_item_checkout_id_fk FOREIGN
KEY (checkout_id) REFERENCES checkout (checkout_id) ON DELETE CASCADE ON
UPDATE CASCADE;
ALTER TABLE supply_item ADD CONSTRAINT supply_item_supply_id_fk FOREIGN KEY
(supply_id) REFERENCES supply (supply_id) ON DELETE CASCADE ON UPDATE
CASCADE;
ALTER TABLE write_off_item ADD CONSTRAINT write_off_item_write_off_id_fk
FOREIGN KEY (write_off_id) REFERENCES write_off (write_off_id) ON DELETE
CASCADE ON UPDATE CASCADE;
ALTER TABLE book_author ADD CONSTRAINT book_author_book_item_id_fk FOREIGN
KEY (book_item_id) REFERENCES book_item (book_item_id) ON DELETE CASCADE ON
UPDATE CASCADE;
ALTER TABLE book_author ADD CONSTRAINT book_author_author_id_fk FOREIGN KEY
(author_id) REFERENCES author (author_id) ON DELETE CASCADE ON UPDATE
CASCADE;

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ALTER TABLE event ADD CONSTRAINT event_employee_id_fk FOREIGN KEY
(employee_id) REFERENCES employee (employee_id) ON DELETE RESTRICT;
ALTER TABLE checkout_item ADD CONSTRAINT checkout_item_book_item_id_fk
FOREIGN KEY (book_item_id) REFERENCES book_item (book_item_id) ON DELETE
RESTRICT;
ALTER TABLE write_off_item ADD CONSTRAINT write_off_item_book_item_id_fk
FOREIGN KEY (book_item_id) REFERENCES book_item (book_item_id) ON DELETE
RESTRICT;
ALTER TABLE supply_item ADD CONSTRAINT supply_item_book_item_id_fk FOREIGN
KEY (book_item_id) REFERENCES book_item (book_item_id) ON DELETE RESTRICT;
```